

A bright yellow lightning bolt strikes diagonally across a dark blue background, creating a sense of energy and power.

19B ELECTROCHEMISTRY

UNIT 03: WHAT REACTIONS PROVIDE

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 19

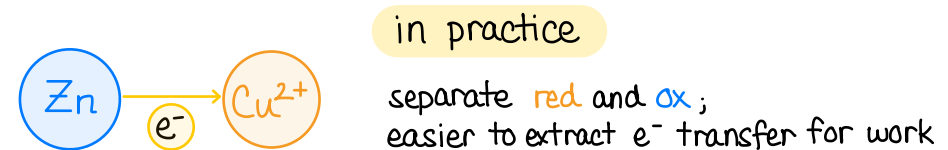
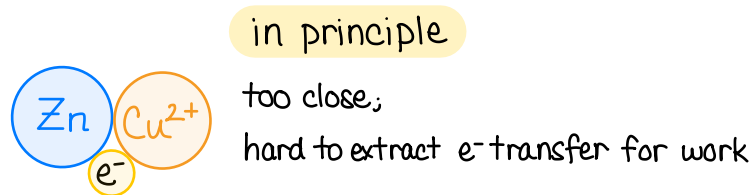
1. Define and assign oxidation states or oxidation numbers.
2. Define aspects of redox reactions: oxidation, reduction, oxidizing agents, and reducing agents.
3. Write half-reactions from overall redox reactions, and vice versa.
4. Balance redox reactions.
5. Describe the components of voltaic (or Galvanic) cells using appropriate terminology and notation.
6. Calculate cell potentials using tables of standard reduction potentials.
7. Compare the strengths of oxidizing and reducing agents using tables of standard reduction potentials.
8. Describe the relationship between the standard cell potential and the equilibrium constant.
9. Perform calculations involving standard cell potentials and equilibrium constants.
10. Calculate the effects of concentration on cell potential using the Nernst equation.
11. Describe how electricity can be used to drive redox reactions.
12. Perform stoichiometric calculations involving mass and current in electrolytic applications.

Voltaic Cells

19.4

Electrochemical cells

A system composed of **two separated half-cells** (corresponding to two half-reactions) that are connected by a salt bridge and a wire to allow for electron transfer.



There are two types:

1) Voltaic cells (also called *Galvanic cells*) $\Delta G_{\text{redox}} < 0 \longrightarrow \text{work}_{\text{electrical}}$

- Use *spontaneous* redox reactions to create electricity (flow of electrons).

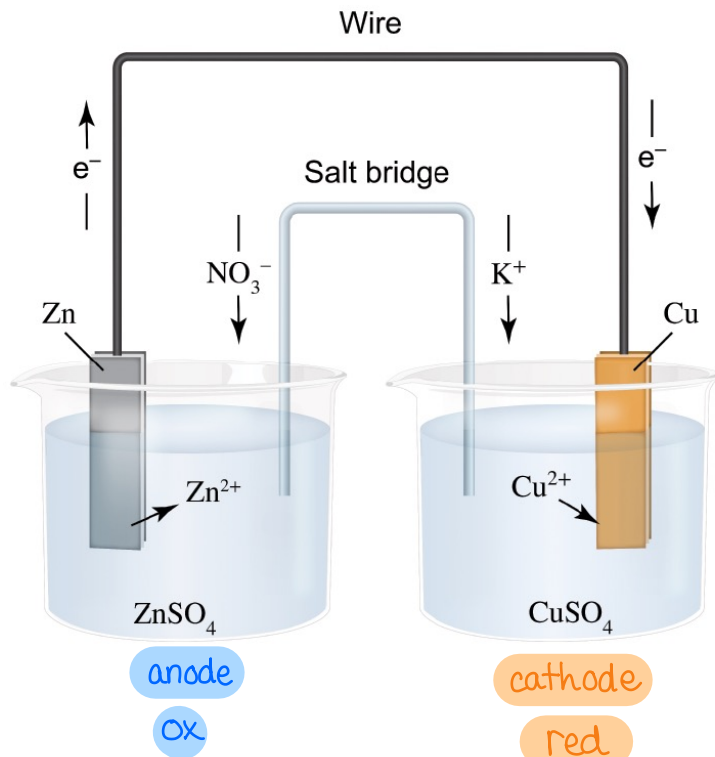
2) Electrolytic cells $\text{work}_{\text{electrical}} \longrightarrow \Delta G_{\text{redox}} > 0$

- Use electricity to cause a *nonspontaneous* redox reaction to occur.

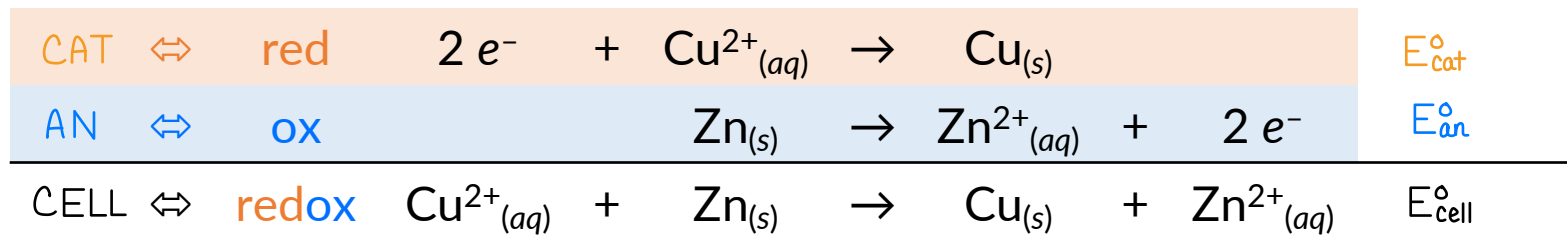
Voltaic Cells

19.4

Voltaic/Galvanic Cell Summary



- The overall redox reaction is *spontaneous*.
- Chemical (free) energy is converted into electrical energy.
 - Oxidation occurs at the anode half-cell.
 - Reduction occurs at the cathode half-cell.
 - Electrons flow from anode to cathode through a wire.
 - Salt bridge anions flow into the anode.
 - Salt bridge cations flow into the cathode.



The sustained flow of electrons produces a current.

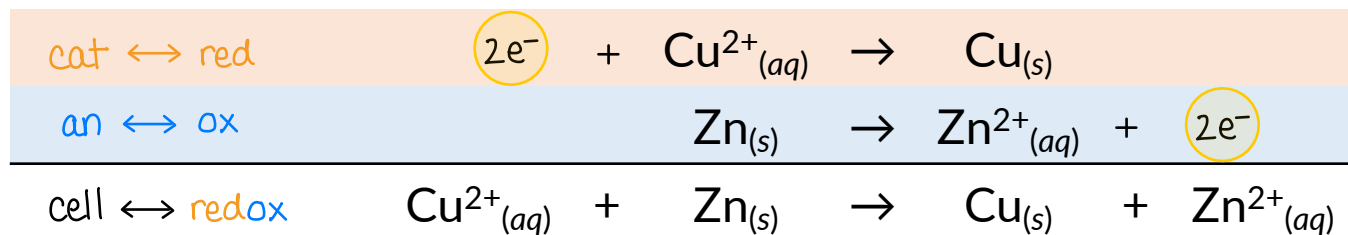
The driving force for the flow of electrons is called the cell potential (E_{cell}), measured in volts (V).

Voltaic Cells

19.4

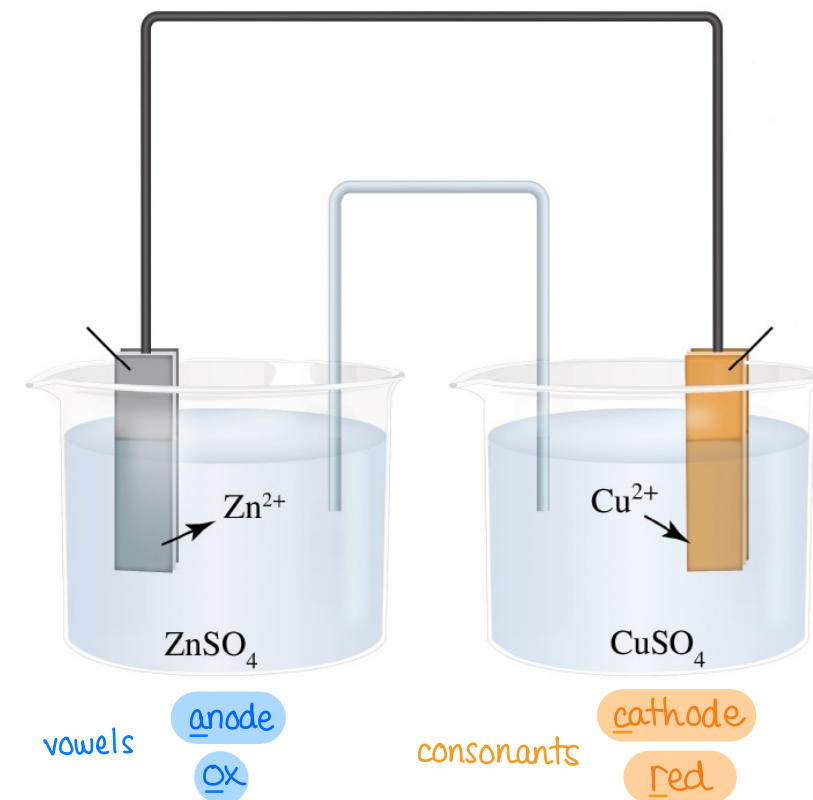
The *spontaneous* redox reaction can occur with the reduction and oxidation half-reactions in *separated* compartments, provided there are *connections* that allow for:

- The flow of electrons (e^-) through electrodes and a wire
- The flow of ions through a salt bridge



Anode half-cell: site of *oxidation* half-reaction; *loss of e^-*

Cathode half-cell: site of *reduction* half-reaction; *gain of e^-*

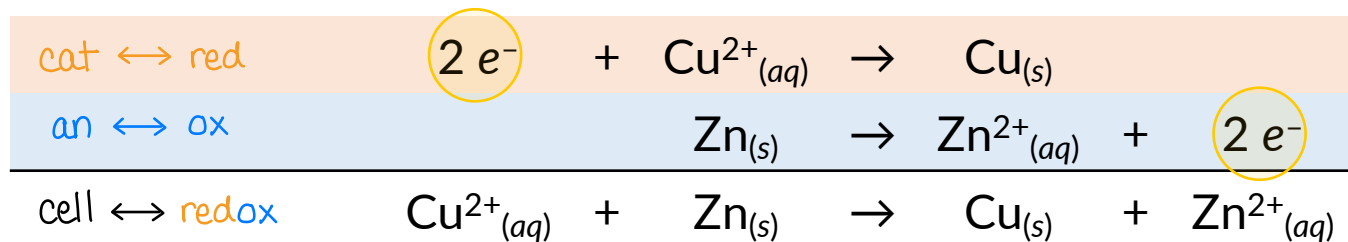


Voltaic Cells

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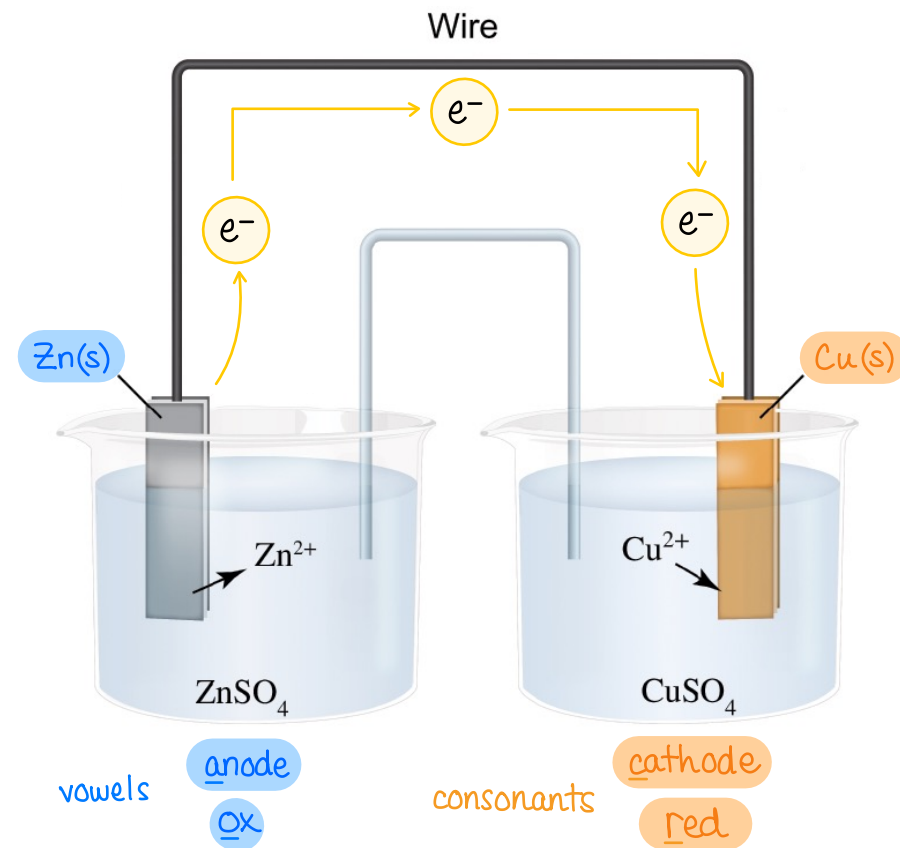
Electrodes: Conductive, solid metal bars in the anode and cathode half-cells.

- Upon **oxidation** of Zn(s) from the Zn electrode, the e^- move through the **wire**, then the **Cu electrode**, and into the cathodic solution of Cu^{2+} ions.
- These e^- **reduce** Cu^{2+} into Cu(s).



Anode half-cell: site of **oxidation** half-reaction; **loss of e^-**

Cathode half-cell: site of **reduction** half-reaction; **gain of e^-**

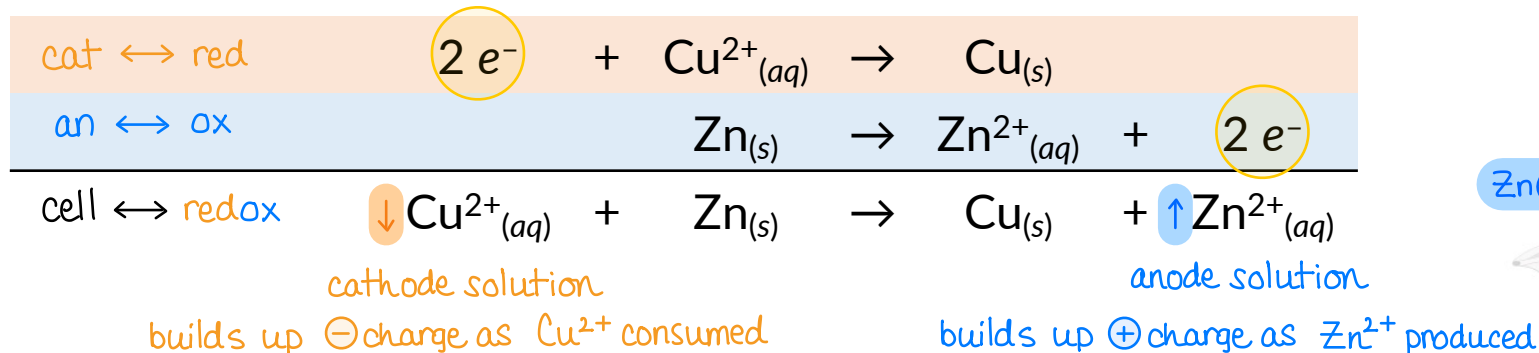


Voltaic Cells

19.4

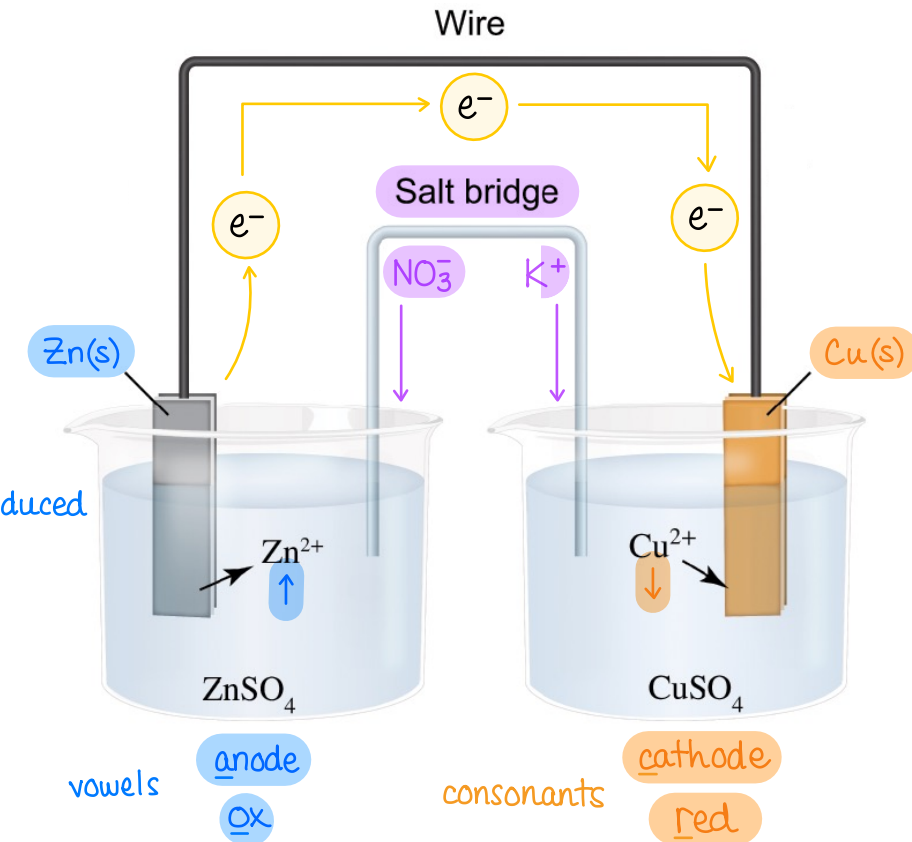
The salt bridge completes the electrical connection between the two half-reactions. It contains an electrolyte solution that offsets/balances the build-up of charge resulting from electron flow.

- Salt **cations** flow into the **cathode** e.g. K^+ if KNO_3 is salt
- Salt **anions** flow into the **anode** e.g. NO_3^- if KNO_3 is salt



Anode half-cell: site of **oxidation** half-reaction; **loss of e^-**

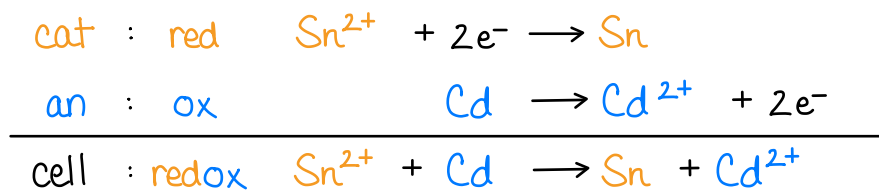
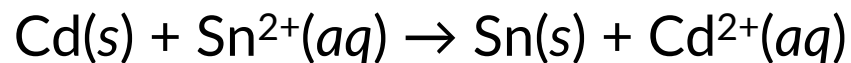
Cathode half-cell: site of **reduction** half-reaction; **gain of e^-**



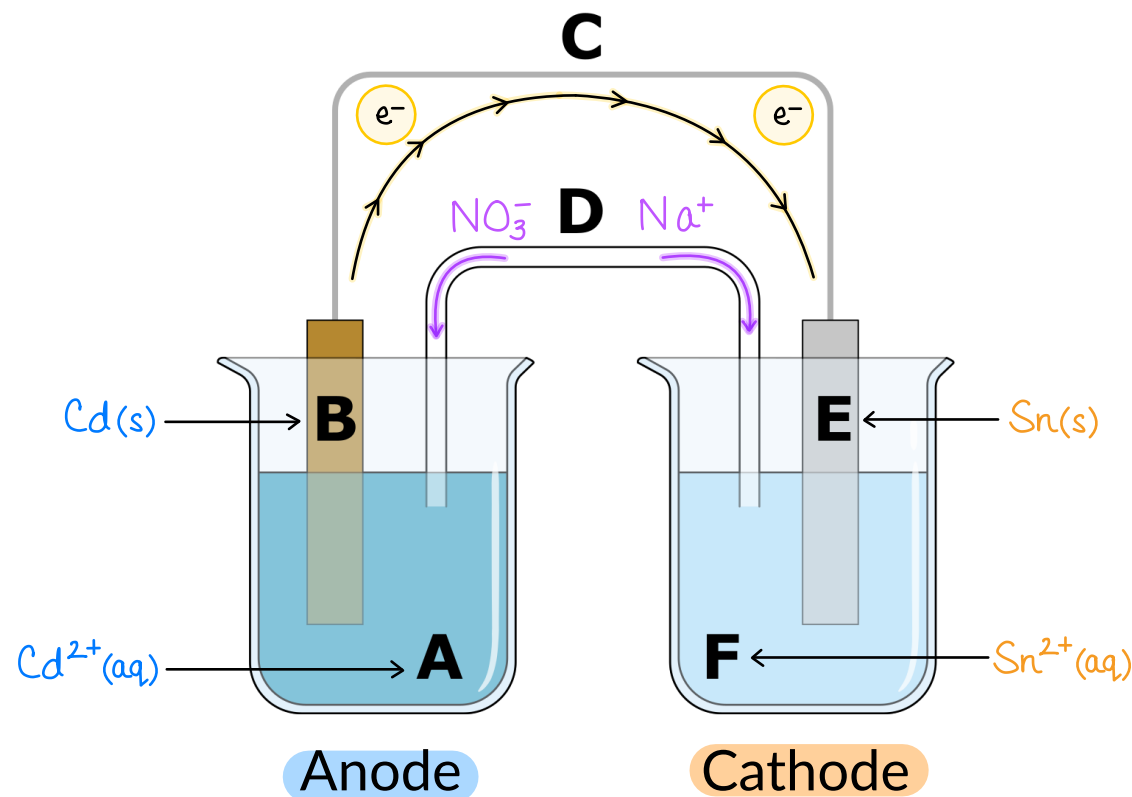
IN-CLASS QUESTION 1

Learning Objective(s): 19.3, 19.5

On the cell diagram, label the following properties based on the spontaneous redox reaction given below. Suppose the salt bridge is comprised of $\text{NaNO}_3(aq)$. $\longrightarrow \text{Na}^+(aq) + \text{NO}_3^-(aq)$
Cation anion



- ☒ The ions in solution (**A** and **F**).
- ☒ The electrode material (**B** and **E**).
- ☒ The flow of electrons across the wire (**C**).
- ☒ The flow of salt ions in the salt bridge (**D**).



Cell Potentials

19.5

Standard cell potential (E°_{cell})

The cell potential (in volts, V) of an electrochemical cell measured under standard conditions.

- Standard denoted by $^\circ$ symbol
- Any aqueous solutes are 1 M.
- Any gases are at 1 atm pressure.
- Any solids and liquids are pure.
- Temperature often 25 $^\circ\text{C}$.

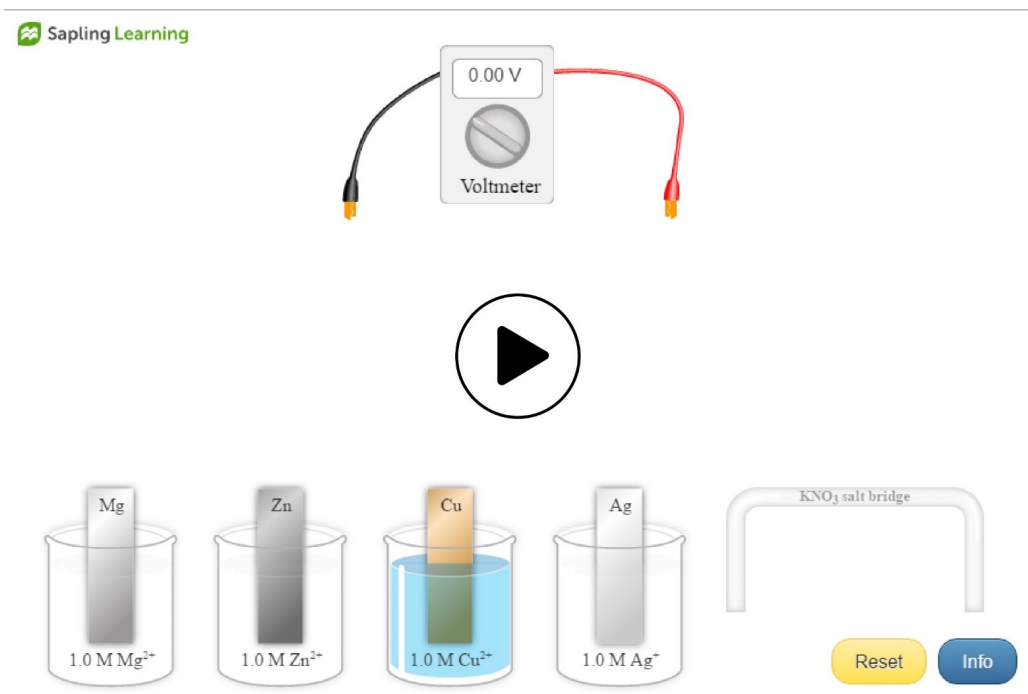


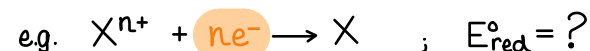
FIGURE 19.8 Interactive in e-book

Cell Potentials

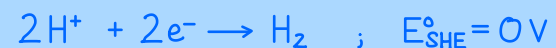
19.5

Even more conventions to follow:

- All half-reactions are written as *reduction* half-reactions with standard *reduction potentials* (E_{red}°).



- The standard hydrogen electrode (SHE) is *arbitrarily* assigned a value of



- When the SHE is paired with *any other* half-cell, the measured potential of the overall cell is assigned to the non-SHE half-cell as its standard reduction potential (E_{red}°).

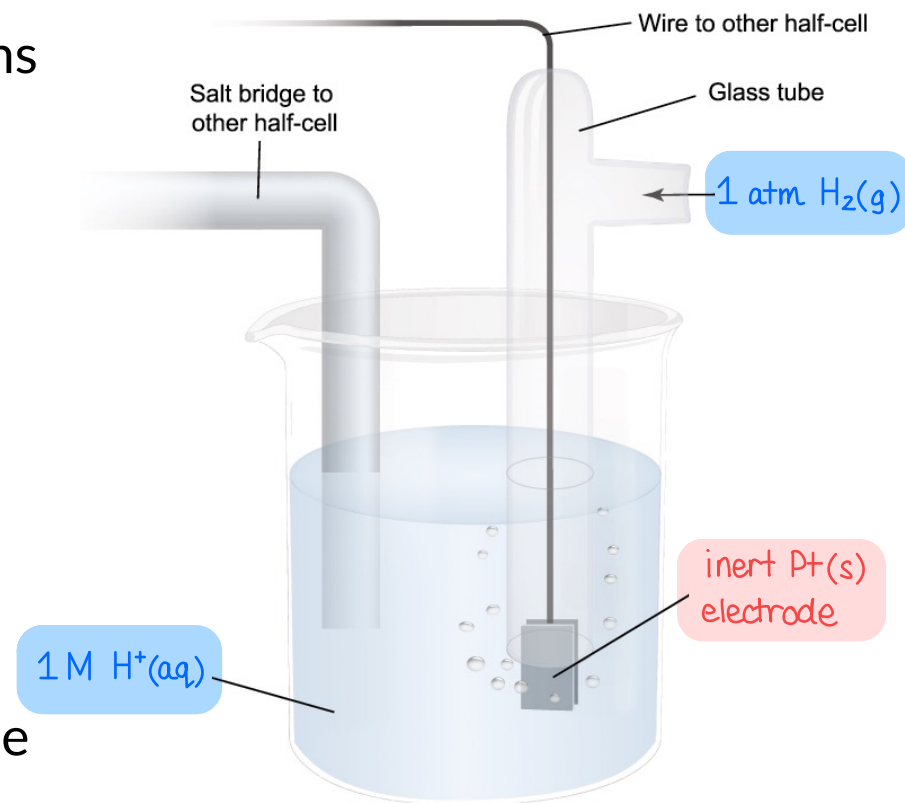
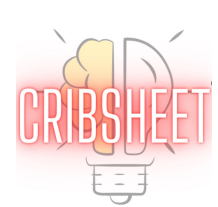


FIGURE 19.7



Cell Potentials

19.5

Tables of standard reduction potentials (E_{red}°) will be provided when needed.

The E_{red}° of any two half-reactions can be combined to calculate any E_{cell}° .

$$E_{\text{cell}}^{\circ} = E_{\text{cat}}^{\circ} - E_{\text{an}}^{\circ}$$

both reduction potentials

A voltaic (Galvanic) will have $E_{\text{cell}}^{\circ} > 0 \text{ V}$.

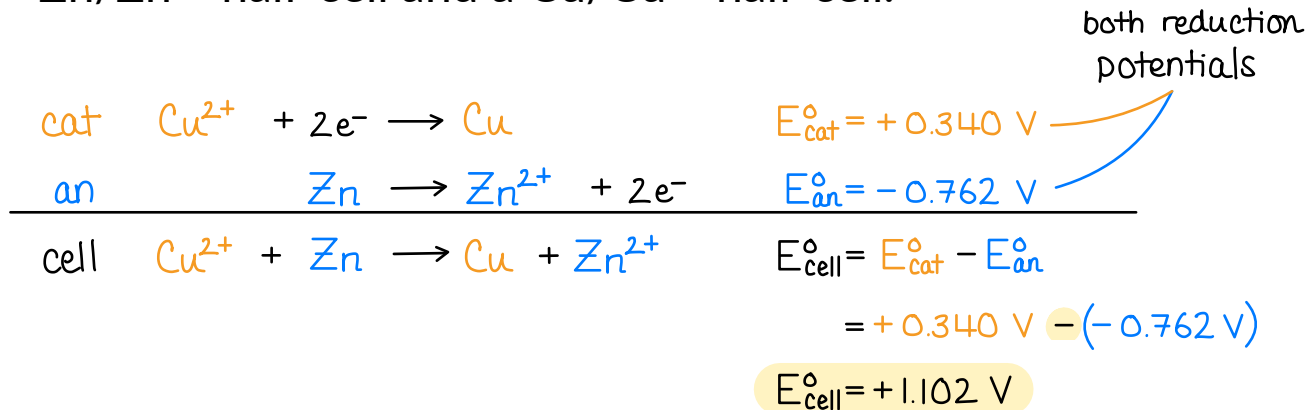
- The **more positive E_{red}°** is assigned as the cathode half-cell ($E_{\text{cathode}}^{\circ}$).
- The **less positive E_{red}°** is assigned as the anode half-cell (E_{anode}°).

Reduction Half-Reaction								E_{red}° (V)
3	e^{-}	+	1	$\text{Au}^{3+} (aq)$	$\rightarrow$	1	$\text{Au} (s)$	+1.498 V
2	e^{-}	+	1	$\text{Cl}_2 (g)$	$\rightarrow$	2	$\text{Cl}^{-} (aq)$	+1.358 V
1	e^{-}	+	1	$\text{Ag}^{+} (aq)$	$\rightarrow$	1	$\text{Ag} (s)$	+0.800 V
1	e^{-}	+	1	$\text{Fe}^{3+} (aq)$	$\rightarrow$	1	$\text{Fe}^{2+} (aq)$	+0.771 V
1	e^{-}	+	1	$\text{Cu}^{+} (aq)$	$\rightarrow$	1	$\text{Cu} (s)$	+0.521 V
2	e^{-}	+	1	$\text{Cu}^{2+} (aq)$	$\rightarrow$	1	$\text{Cu} (s)$	+0.340 V
2	e^{-}	+	2	$\text{H}^{+} (aq)$	$\rightarrow$	1	$\text{H}_2 (g)$	SHE 0 V
2	e^{-}	+	1	$\text{Pb}^{2+} (aq)$	$\rightarrow$	1	$\text{Pb} (s)$	−0.126 V
2	e^{-}	+	1	$\text{Sn}^{2+} (aq)$	$\rightarrow$	1	$\text{Sn} (s)$	−0.138 V
2	e^{-}	+	1	$\text{Ni}^{2+} (aq)$	$\rightarrow$	1	$\text{Ni} (s)$	−0.257 V
2	e^{-}	+	1	$\text{Cd}^{2+} (aq)$	$\rightarrow$	1	$\text{Cd} (s)$	−0.403 V
2	e^{-}	+	1	$\text{Fe}^{2+} (aq)$	$\rightarrow$	1	$\text{Fe} (s)$	−0.447 V
3	e^{-}	+	1	$\text{Cr}^{3+} (aq)$	$\rightarrow$	1	$\text{Cr} (s)$	−0.744 V
2	e^{-}	+	1	$\text{Zn}^{2+} (aq)$	$\rightarrow$	1	$\text{Zn} (s)$	−0.762 V
3	e^{-}	+	1	$\text{Al}^{3+} (aq)$	$\rightarrow$	1	$\text{Al} (s)$	−1.662 V
2	e^{-}	+	1	$\text{Mg}^{2+} (aq)$	$\rightarrow$	1	$\text{Mg} (s)$	−2.372 V
1	e^{-}	+	1	$\text{Na}^{+} (aq)$	$\rightarrow$	1	$\text{Na} (s)$	−2.710 V

EXAMPLE PROBLEM

Learning Objective(s): 19.3, 19.4, 19.6

Calculate the standard cell potential (E_{cell}° in V) of the Daniell cell, which is a voltaic cell consisting of a Zn/Zn²⁺ half-cell and a Cu/Cu²⁺ half-cell.



- notes
- more positive E_{red}° is cathode
 - don't "flip" sign of E_{an}°

Reduction Half-Reaction							E_{red}° (V)	
3	e ⁻	+	1	Au ³⁺ (aq)	→	1	Au (s)	+1.498 V
2	e ⁻	+	1	Cl ₂ (g)	→	2	Cl ⁻ (aq)	+1.358 V
1	e ⁻	+	1	Ag ⁺ (aq)	→	1	Ag (s)	+0.800 V
1	e ⁻	+	1	Fe ³⁺ (aq)	→	1	Fe ²⁺ (aq)	+0.771 V
1	e ⁻	+	1	Cu ⁺ (aq)	→	1	Cu (s)	+0.521 V
2	e ⁻	+	1	Cu ²⁺ (aq)	→	1	Cu (s)	+0.340 V
2	e ⁻	+	2	H ⁺ (aq)	→	1	H ₂ (g)	0 V
2	e ⁻	+	1	Pb ²⁺ (aq)	→	1	Pb (s)	-0.126 V
2	e ⁻	+	1	Sn ²⁺ (aq)	→	1	Sn (s)	-0.138 V
2	e ⁻	+	1	Ni ²⁺ (aq)	→	1	Ni (s)	-0.257 V
2	e ⁻	+	1	Cd ²⁺ (aq)	→	1	Cd (s)	-0.403 V
2	e ⁻	+	1	Fe ²⁺ (aq)	→	1	Fe (s)	-0.447 V
3	e ⁻	+	1	Cr ³⁺ (aq)	→	1	Cr (s)	-0.744 V
2	e ⁻	+	1	Zn ²⁺ (aq)	→	1	Zn (s)	-0.762 V
3	e ⁻	+	1	Al ³⁺ (aq)	→	1	Al (s)	-1.662 V
2	e ⁻	+	1	Mg ²⁺ (aq)	→	1	Mg (s)	-2.372 V
1	e ⁻	+	1	Na ⁺ (aq)	→	1	Na (s)	-2.710 V

How old are batteries?

Chemistry in Context

The Daniell cell (middle) was used as the definition for “volt” until the 1880s.



John Frederic Daniell (left)
and Michael Faraday (right)

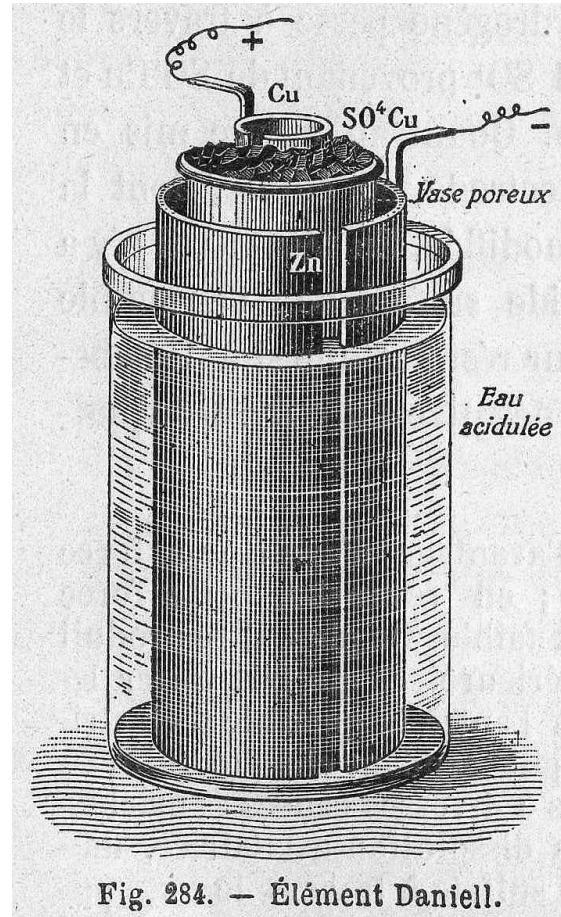


Fig. 284. — Élément Daniell.

Let's build a “button” battery!

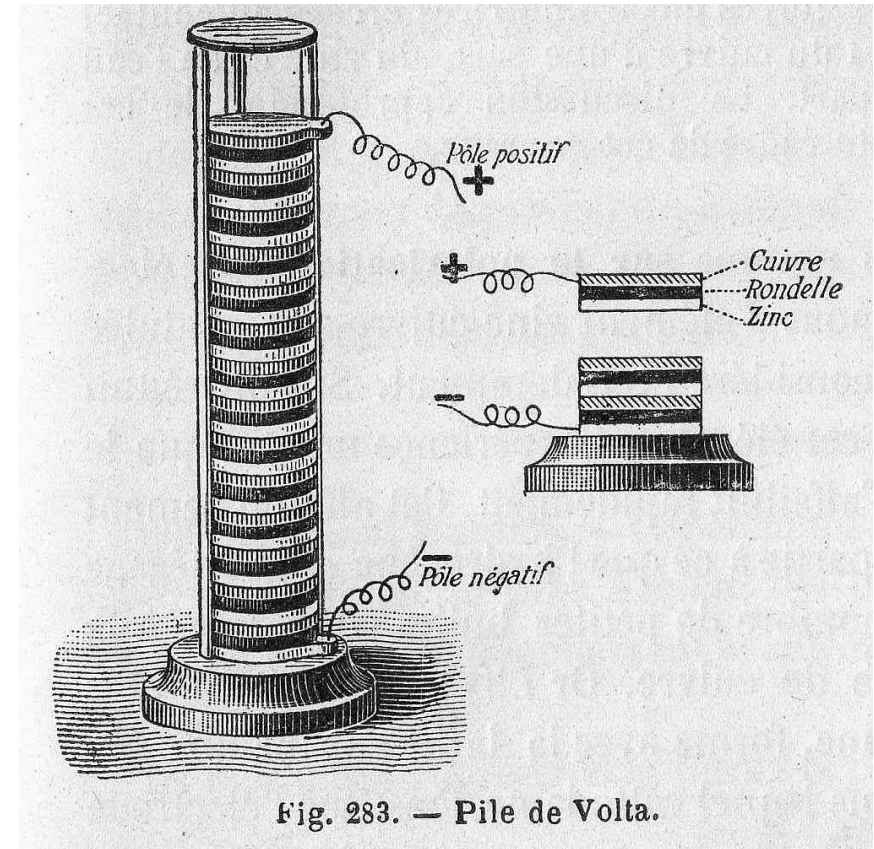


Fig. 283. — Pile de Volta.

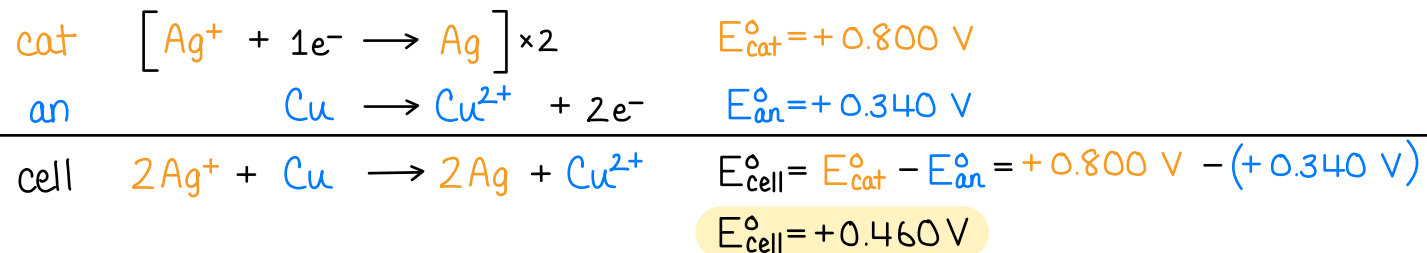
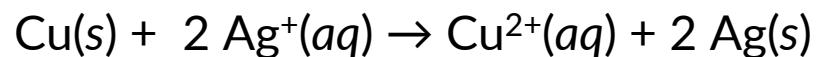
EXAMPLE PROBLEM

Learning Objective(s): 19.3, 19.4, 19.6

When the half-reactions are multiplied by integers to balance the overall redox/cell reaction, the value of E_{cell}° does NOT change.

- In other words, do NOT multiply the standard reduction potential (E_{red}°) by the integers.

Calculate the standard cell potential (E_{cell}° in V) for the following reaction.



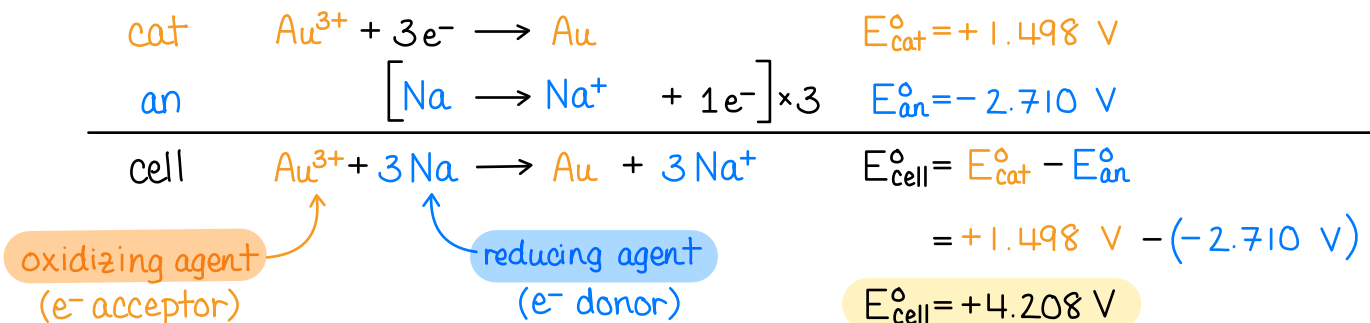
Reduction Half-Reaction								E_{red}° (V)
3	e^{-}	+	1	$\text{Au}^{3+}(\text{aq})$	$\rightarrow$	1	$\text{Au}(\text{s})$	+1.498 V
2	e^{-}	+	1	$\text{Cl}_2(\text{g})$	$\rightarrow$	2	$\text{Cl}^{-}(\text{aq})$	+1.358 V
1	e^{-}	+	1	$\text{Ag}^{+}(\text{aq})$	$\rightarrow$	1	$\text{Ag}(\text{s})$	+0.800 V
1	e^{-}	+	1	$\text{Fe}^{3+}(\text{aq})$	$\rightarrow$	1	$\text{Fe}^{2+}(\text{aq})$	+0.771 V
1	e^{-}	+	1	$\text{Cu}^{+}(\text{aq})$	$\rightarrow$	1	$\text{Cu}(\text{s})$	+0.521 V
2	e^{-}	+	1	$\text{Cu}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Cu}(\text{s})$	+0.340 V
2	e^{-}	+	2	$\text{H}^{+}(\text{aq})$	$\rightarrow$	1	$\text{H}_2(\text{g})$	0 V
2	e^{-}	+	1	$\text{Pb}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Pb}(\text{s})$	−0.126 V
2	e^{-}	+	1	$\text{Sn}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Sn}(\text{s})$	−0.138 V
2	e^{-}	+	1	$\text{Ni}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Ni}(\text{s})$	−0.257 V
2	e^{-}	+	1	$\text{Cd}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Cd}(\text{s})$	−0.403 V
2	e^{-}	+	1	$\text{Fe}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Fe}(\text{s})$	−0.447 V
3	e^{-}	+	1	$\text{Cr}^{3+}(\text{aq})$	$\rightarrow$	1	$\text{Cr}(\text{s})$	−0.744 V
2	e^{-}	+	1	$\text{Zn}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Zn}(\text{s})$	−0.762 V
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2	e^{-}	+	1	$\text{Mg}^{2+}(\text{aq})$	$\rightarrow$	1	$\text{Mg}(\text{s})$	−2.372 V
1	e^{-}	+	1	$\text{Na}^{+}(\text{aq})$	$\rightarrow$	1	$\text{Na}(\text{s})$	−2.710 V

IN-CLASS QUESTION 2

Learning Objective(s): 19.3, 19.4, 19.6

Using the table of standard reduction potentials given, construct a Galvanic cell with the **largest E°_{cell}** value.

a) Write the balanced redox reaction.



b) Identify the reducing and oxidizing agents.

c) How many electrons (n) are transferred? $n=3$

d) Calculate E°_{cell} .

Reduction Half-Reaction								E°_{red} (V)
3	e^-	+	1	$\text{Au}^{3+} (\text{aq})$	$\rightarrow$	1	$\text{Au} (\text{s})$	+1.498 V
2	e^-	+	1	$\text{Cl}_2 (\text{g})$	$\rightarrow$	2	$\text{Cl}^- (\text{aq})$	+1.358 V
1	e^-	+	1	$\text{Ag}^+ (\text{aq})$	$\rightarrow$	1	$\text{Ag} (\text{s})$	+0.800 V
1	e^-	+	1	$\text{Fe}^{3+} (\text{aq})$	$\rightarrow$	1	$\text{Fe}^{2+} (\text{aq})$	+0.771 V
1	e^-	+	1	$\text{Cu}^+ (\text{aq})$	$\rightarrow$	1	$\text{Cu} (\text{s})$	+0.521 V
2	e^-	+	1	$\text{Cu}^{2+} (\text{aq})$	$\rightarrow$	1	$\text{Cu} (\text{s})$	+0.340 V
2	e^-	+	2	$\text{H}^+ (\text{aq})$	$\rightarrow$	1	$\text{H}_2 (\text{g})$	0 V
2	e^-	+	1	$\text{Pb}^{2+} (\text{aq})$	$\rightarrow$	1	$\text{Pb} (\text{s})$	−0.126 V
2	e^-	+	1	$\text{Sn}^{2+} (\text{aq})$	$\rightarrow$	1	$\text{Sn} (\text{s})$	−0.138 V
2	e^-	+	1	$\text{Ni}^{2+} (\text{aq})$	$\rightarrow$	1	$\text{Ni} (\text{s})$	−0.257 V
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2	e^-	+	1	$\text{Mg}^{2+} (\text{aq})$	$\rightarrow$	1	$\text{Mg} (\text{s})$	−2.372 V
1	e^-	+	1	$\text{Na}^+ (\text{aq})$	$\rightarrow$	1	$\text{Na} (\text{s})$	−2.710 V

BONUS QUESTION

Assessed on: Final Exam, Summer 2023

Calculate E_{cell}° (in volts, V) for a Galvanic cell that uses the Cl_2/Cl^- and Ga^{3+}/Ga half-cells.

Then, determine how many electrons (n) are transferred?

$$\text{Cl}_2/\text{Cl}^- \quad E_{\text{red}}^{\circ} = +1.36 \text{ V}$$

$$\text{Ga}^{3+}/\text{Ga} \quad E_{\text{red}}^{\circ} = -0.55 \text{ V}$$